

variation in ozone levels and mortality counts, results have been inconclusive. The authors investigated whether short-term (daily and weekly) exposure to ambient ozone is associated with mortality in the United States. Using analytical methods and databases developed for the National Morbidity, Mortality, and Air Pollution Study, they estimated a national average relative rate of mortality associated with short-term exposure to ambient ozone for 95 large US urban communities from 1987-2000. The authors used distributed-lag models for estimating community-specific relative rates of mortality adjusted for time-varying confounders (particulate matter, weather, seasonality, and long-term trends) and hierarchical models for combining relative rates across communities to estimate a national average relative rate, taking into account spatial heterogeneity. A 10-ppb increase in the previous week's ozone was associated with a 0.52% increase in daily mortality (95% posterior interval [PI], 0.27%-0.77%) and a 0.64% increase in cardiovascular and respiratory mortality (95% PI, 0.31%-0.98%). Effect estimates for aggregate ozone during the previous week were larger than for models considering only a single day's exposure. Results were robust to adjustment for particulate matter, weather, seasonality, and long-term trends. These results indicate a statistically significant association between short-term changes in ozone and mortality on average for 95 large US urban communities, which include about 40% of the total US population.

Non-Accidental Mortality

The coefficient and standard error are based on the relative risk (1.003908) and 95% confidence interval (1.0013-1.0065) associated with a 10 ppb increase in daily average ozone (Bell et al., 2004, p. 2376).

G.2.1.2 Bell et al. (2005)

Although many time-series studies of ozone and mortality have identified positive associations, others have yielded null or inconclusive results, making the results of these studies difficult to interpret. The authors performed a meta-analysis of 144 effect estimates from 39 time-series studies, and estimated pooled effects by lags, age groups, cause-specific mortality, and concentration metrics. They compared results with pooled estimates from the National Morbidity, Mortality, and Air Pollution Study (NMMAPS), a time-series study of 95 large U.S. urban centers from 1987 to 2000. Both meta-analysis and NMMAPS results provided strong evidence of a short-term association between ozone and mortality, with larger effects for cardiovascular and respiratory mortality, the elderly, and current-day ozone exposure. In both analyses, results were insensitive to adjustment for particulate matter and model specifications. In the meta-analysis, a 10-ppb increase in daily ozone at single-day or 2-day average of lags 0, 1, or 2 days was associated with an 0.87% increase in total mortality (95% posterior interval = 0.55% to 1.18%), whereas the lag 0 NMMAPS estimate is 0.25% (0.12% to 0.39%). Several findings indicate possible publication bias: meta-analysis results were consistently larger than those from NMMAPS; meta-analysis pooled estimates at lags 0 or 1 were larger when only a single lag was reported than when estimates for multiple lags were reported; and heterogeneity of city-specific estimates in the meta-analysis were larger than with NMMAPS.

All-Cause Mortality

The coefficient and standard error are based on the relative risk (1.008738) and 95% confidence interval (1.0055-1.0119) associated with a 10 ppb increase in daily average ozone (Bell et al., 2005, Table 6).

G.2.1.3 Huang et al. (2005)

The authors developed Bayesian hierarchical distributed lag models for estimating associations between daily variations in summer ozone levels and daily variations in cardiovascular and respiratory (CVDRESP) mortality counts for 19 large U.S. cities included in the National Morbidity, Mortality and Air Pollution Study (NMMAPS) for the summers of 1987-1994. In the first stage, they defined a semi-parametric distributed lag Poisson regression model to estimate city-specific relative rates of CVDRESP mortality associated with short-term exposure to summer ozone. In the second stage, they specified a class of distributions for the true city-specific relative rates to estimate an overall effect by taking into account the variability within and across cities. They performed the calculations with respect to several random effects distributions (normal, t-student, and mixture of normal), thus relaxing the common assumption of a two-stage normal-normal hierarchical model. They assessed the sensitivity of the results to: (i) lag structure for ozone exposure; (ii) degree of adjustment for long-term trends; (iii) inclusion of other pollutants in the model; (iv) heat waves; (v) random effects distributions; and (vi) prior hyperparameters. On average across cities, the authors found that a 10 ppb increase in summer ozone level over the previous week is associated with a 1.25 per cent increase in CVDRESP mortality (95 per cent posterior regions: 0.47, 2.03). The relative rate estimates are also positive and statistically significant at lags 0, 1 and 2. They found that associations between summer ozone and CVDRESP mortality are sensitive to the confounding adjustment for PM₁₀, but are robust to: (i) the adjustment for long-term trends, other gaseous pollutants (NO₂, SO₂ and CO); (ii) the distributional assumptions at the second stage of the hierarchical model; and (iii) the prior distributions on all unknown parameters.

Cardiopulmonary Mortality

Assuming a 10 ppb change in ozone, Huang et al. (2005, Table 1) reported a 1.25% change in CVDRESP mortality with a 95% confidence interval of 0.47% to 2.03%.

Note that Huang et al. (2005, p. 549) define CVDRESP as including ICD-9 codes: 390-448, 480-487, 490-496, and 507. This differs somewhat from the the definition of “cardiopulmonary” mortality in BenMAP -- defined as ICD-9 codes 401-440 and 460-519.

G.2.1.4 Ito and Thurston (1996)

In this study, race, gender, and cause-specific counts of daily mortality in Cook County, Illinois (which encompasses the city of Chicago) during 1985-1990 were analyzed to determine if there was any heterogeneity in air pollution/weather/mortality associations across these various population subcategories. Seasonal cross-correlations

between mortality and environmental variables first were examined to identify appropriate lag structures. Of the pollution variables considered -- PM₁₀, ozone, CO, SO₂, and visual range-derived extinction coefficient -- both PM₁₀ and ozone showed significant associations with same-day and next-day mortality. The Poisson regression models employed included seasonal cycles (sine/cosine series), square and linear terms of lagged temperature, trend line, day-of-week dummy variables, and the average of the same day's and previous day's PM₁₀ or ozone.

The authors reported a significant relationship for ozone and PM₁₀ with both pollutants in the model; no significant effects were found for SO₂ and CO. In single pollutant models the effects were slightly larger. The health impact function for ozone is based on results from the co-pollutant models.

Non-Accidental Mortality

For a co-pollutant model with PM₁₀, the ozone coefficient (0.000634) and standard error (0.000251) were obtained directly from the author because the published paper reported incorrect information.

G.2.1.5 Ito et al. (2005)

The authors conducted a review and meta-analysis of short-term ozone mortality studies, identified unresolved issues, and conducted an additional time-series analysis for 7 U.S. cities (Chicago, Detroit, Houston, Minneapolis-St. Paul, New York City, Philadelphia, and St. Louis). They found a combined estimate of 0.39% (95% confidence interval = 0.26-0.51%) per 10-ppb increase in 1-hour daily maximum ozone for the all-age nonaccidental cause/single pollutant model (43 studies). Adjusting for the funnel plot asymmetry resulted in a slightly reduced estimate (0.35%; 0.23-0.47%). In a subset for which particulate matter (PM) data were available (15 studies), the corresponding estimates were 0.40% (0.27-0.53%) for ozone alone and 0.37% (0.20-0.54%) with PM in model. The estimates for warm seasons were generally larger than those for cold seasons. The additional time-series analysis found that including PM in the model did not substantially reduce the ozone risk estimates. However, the difference in the weather adjustment model could result in a 2-fold difference in risk estimates (eg, 0.24% to 0.49% in multicity combined estimates across alternative weather models for the ozone-only all-year case). The authors concluded that the results suggest short-term associations between ozone and daily mortality in the majority of the cities, although the estimates appear to be heterogeneous across cities.

Non-Accidental Mortality

Ito et al. (2005) reported results for functions with 1-hour daily maximum, 24-hour daily average metrics, and 8-hour maximum from the 24-hour mean. We include the warm season 8-hour maximum from the 24-hour mean. Assuming a 20 ppb change in the daily 24-hour average, Ito et al. (2005, p. 448) reported a 3.5% change in non-accidental mortality with a 95% confidence interval of 2.1% to 4.9%.

One-hour Max Function

Assuming a 10 ppb change in the daily 1-hour maximum, Ito et al. (2005, p. 446) reported a 0.40% change in non-accidental mortality with a 95% confidence interval of 0.27% to 0.53%.

Daily Average Function

Assuming a 20 ppb change in the daily 24-hour average, Ito et al. (2005, p. 448) reported a 3.5% change in non-accidental mortality with a 95% confidence interval of 2.1% to 4.9%.

G.2.1.6 Jerrett et al. (2009)

Jerrett et al. (2009) examined the potential contribution of long-term ozone exposure to the risk of death from cardiopulmonary causes and specifically to death from respiratory causes. Data from the study cohort of the American Cancer Society Cancer Prevention Study II were correlated with air-pollution data from 96 metropolitan statistical areas in the United States. Associations between ozone concentrations and the risk of death were evaluated with the use of standard and multilevel Cox regression models. In single-pollutant models, increased concentrations of either PM_{2.5} or ozone were significantly associated with an increased risk of death from cardiopulmonary causes. In two-pollutant models, PM_{2.5} was associated with the risk of death from cardiovascular causes, whereas ozone was associated with the risk of death from respiratory causes. The estimated relative risk of death from respiratory causes that was associated with an increment in ozone concentration of 10 ppb was 1.040 (95% confidence interval, 1.010 to 1.067). The association of ozone with the risk of death from respiratory causes was insensitive to adjustment for confounders and to the type of statistical model used. The authors concluded that they were not able to detect an effect of ozone on the risk of death from cardiovascular causes when the concentration of PM_{2.5} was taken into account. But they did demonstrate a significant increase in the risk of death from respiratory causes in association with an increase in ozone concentration.

Mortality, Respiratory (ICD-9 code 460-519) - 86 U.S. urban areas

In a two-pollutant model the coefficient and standard error are estimated from the relative risk (1.040) and 95% confidence interval (95% CI: 1.013-1.067) for a 10 ppb increase in ambient ozone concentration measured from April to September during the years from 1977 to 2000 in 86 MSAs (Jerrett, et al., 2009, Table 3).

Mortality, Respiratory (ICD-9 code 460-519) - by region

In single-pollutant models the coefficient and standard error for different regions are estimated from the relative risks and 95% confidence intervals for a 10 ppb increase in ambient ozone concentration measured from April to September during the years from 1977 to 2000 (Jerrett, et al., 2009, Table 4).